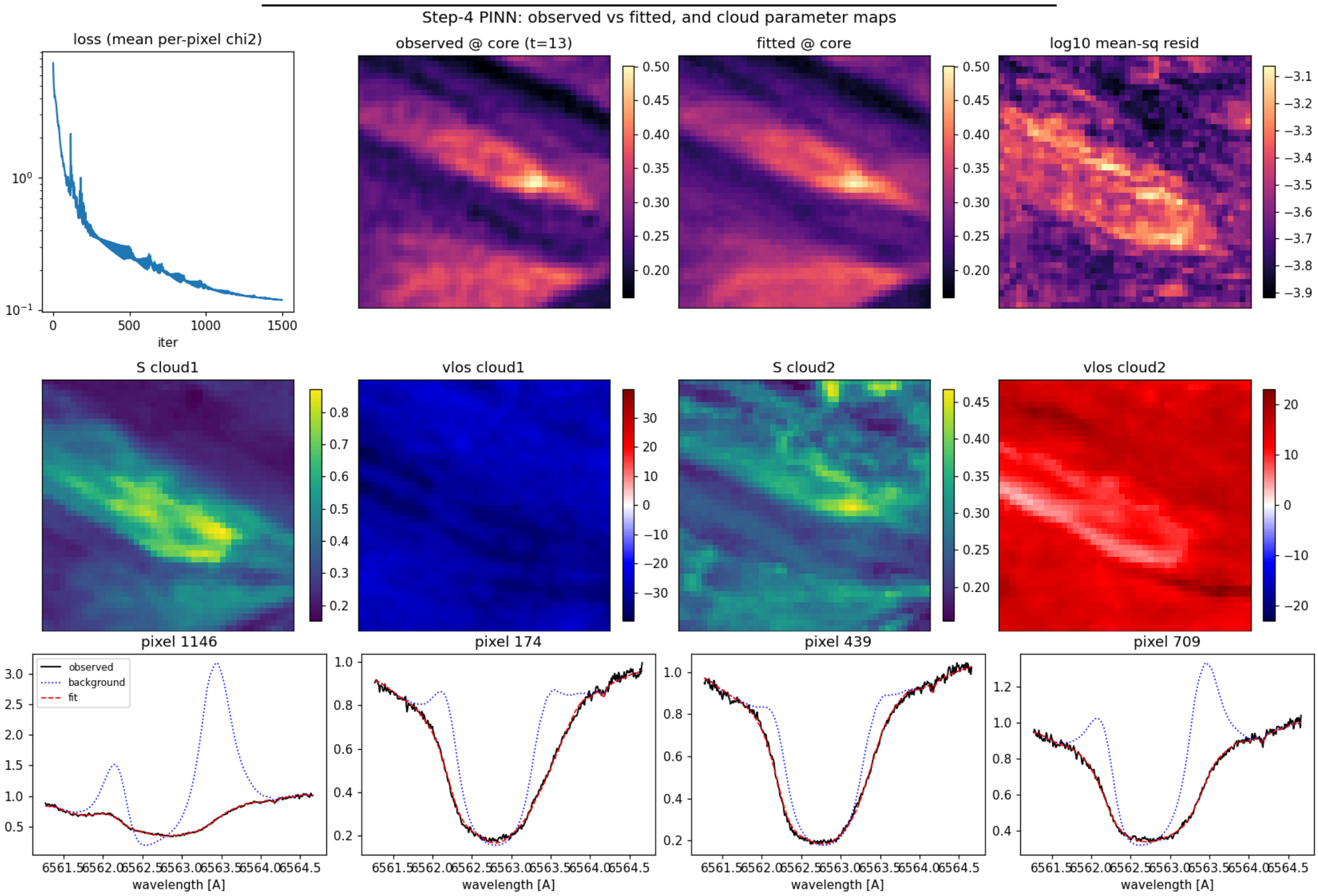
cloudtorch / develop — Step 4 results (PINN)

Space-time MLP + Fourier neural field trained through the two-cloud physics, on the real MiHI Hα cube (`ml` env, CPU). Parameters $= f_\theta(x, y, t)$; `log a` frozen; ranges via sigmoid/softplus; background from PCA (K=6).

Best fit. crop 44×44, 5 frames, $\sigma_{xy} = 4$, $\sigma_t = 2$, width 256, 1500 iters $\rightarrow$ **RMS residual 2.74% of continuum**, mean per-pixel χ^2 0.12 (p95 0.22), 1182 s on CPU. Parameter ranges stayed physical (S [0.151, 0.879], $v_{\mathrm{los},1}$ [-45.9, -20.8] km/s, Δv [11.3, 27.6] km/s); background sits above vs. observed core.

run	σ_{xy}	σ_t	lr	iters	RMS%	χ^2	TV(v_1)	s
`s1_lr3e3_sx4`	4	2	3e-03	700	4.64	0.32	1.81	133
`s2_lr1e2_sx4`	4	2	1e-02	700	2.99	0.13	2.75	130
`s3_lr5e3_sx2`	2	2	5e-03	700	3.25	0.16	1.98	120
`s4_lr5e3_sx6`	6	2	5e-03	700	3.54	0.19	2.43	115
`s5_lr5e3_st1`	4	1	5e-03	700	3.53	0.19	2.19	102
`s6_lr5e3_nf256`	4	2	5e-03	700	3.24	0.16	2.24	105
`s7_lr5e3_conv1500`	4	2	5e-03	1500	2.95	0.13	2.78	220



Takeaways. Fit tracks the data everywhere (RMS 2.7%) and reproduces the filament threads in the core image; residuals concentrate only on the brightest thread. Maps are smooth and physical (S_1 elevated along the filament; $v_{\mathrm{los},2}$ a coherent redshifted field with a ~0 band on the thread; $v_{\mathrm{los},1}$ nearly uniform, i.e. weakly constrained spatially). On average the background sits above the observed core so the clouds carry the absorption, though a minority of pixels show residual background/cloud degeneracy. Sweet spot $\sigma_{xy}{\approx}4$, $\sigma_t{=}2$, lr $5{\times}10^{-3}$, $\gtrsim$1500 iters; the Fourier bandwidths are the space–time regulariser. The physics forward dominates CPU cost (~0.8 s/iter here) – the identical code scales to the full cube on GPU (step 5).